8:59 0.00 KB/s 4G

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main.c... Output

```cpp
#include <iostream>
#include <string>
using namespace std;
// --- Function Prototypes ---
void showDeviceInfo();
void displayVitals(int patientId, float heartRate, float temperature);
int isFeverish(float temperature); // Prototype required by task 1.1.4
int main() {
    // 1.1.1 (a) Void Function - No Parameters
    showDeviceInfo();
    cout << endl;
    // 1.1.2 (b) Void Function - With Parameters
    displayVitals(1, 118, 101.2);
    cout << endl;
    // 1.1.3 & 1.1.4 (c & d) Value-Returning Function & Prototype
    float temp = 101.2;
    if (isFeverish(temp) == 1) {
        cout << "Fever detected!" << endl;
    } else {
        cout << "Temperature n
```

Run

9:00

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main.c... Output

```cpp
            cout << "Temperature normal." << endl;
        }
        return 0;
}
// 1.1.1: Prints the device's name and model
void showDeviceInfo() {
        cout << "Device: VitalCheck Monitor - Model VC-200" << endl;
}

// 1.1.2: Prints a patient's vitals
void displayVitals(int patientId, float heartRate, float temperature) {
        cout << "Patient " << patientId << " -> Heart Rate: " << heartRate
             << " bpm, Temperature: " << temperature << " F" << endl;
}

// 1.1.3 & 1.1.4: Returns 1 if temperature > 100.4, else 0
int isFeverish(float temperatu
        if (temperature > 100.4) {
```

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main.c... Output

```
void showDeviceInfo() {
    cout << "Device: VitalCheck Monitor - Model VC-200" << endl;
}

// 1.1.2: Prints a patient's vitals
void displayVitals(int patientId, float heartRate, float temperature) {
    cout << "Patient " << patientId << " -> Heart Rate: " << heartRate
      << " bpm, Temperature: " << temperature << " F" << endl;
}

// 1.1.3 & 1.1.4: Returns 1 if temperature > 100.4, else 0
int isFeverish(float temperature) {
    if (temperature > 100.4) {
        return 1;
    } else {
        return 0;
    }
}

```

Run

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main.c... Output

```
Device: VitalCheck Monitor - Model VC-200

Patient 1 -> Heart Rate: 118 bpm,
    Temperature: 101.2 F

Fever detected!


=== Code Execution Successful ===
```

9:04

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main.c... Output

```cpp
#include <iostream>
#include <string>
#include <cmath> // Required for pow() in task 1.1.6
using namespace std;
// --- Function Prototypes ---
float calculateBMI(float weightKg, float heightM);
int classifyBMI(float bmi);
int main() {
    // 1.1.5 & 1.1.6 (a & b Home Task) BMI Calculation using pow()
    float weight = 70.0;
    float height = 1.75;
    float bmi = calculateBMI(weight, height);
    cout << "BMI: " << bmi << endl;
    // 1.1.7 (c Home Task) Value-Returning Function – Decision Making
    int categoryCode = classifyBMI(bmi);
    cout << "Category: ";
    if (categoryCode == 1) {
        cout << "Underweight" << endl;
    } else if (categoryCode == 2) {
        cout << "Normal weight" << endl;
```

Run

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main.c... Output

```cpp
    } else if (categoryCode == 3) {
        cout << "Overweight" << endl;
    } else if (categoryCode == 4) {
        cout << "Obese" << endl;
    }
    return 0;
}
// 1.1.5 & 1.1.6: Calculates BMI using pow() library function
float calculateBMI(float weightKg, float heightM) {
    return weightKg / pow(heightM, 2);
}

// 1.1.7: Classifies BMI into numerical categories
int classifyBMI(float bmi) {
    if (bmi < 18.5) {
        return 1; // Underweight
    } else if (bmi >= 18.5 && bmi <= 24.9) {
        return 2; // Normal weight
    } else if (bmi >= 25.0 && bmi <= 29.9) {
        return 3; // Overweight
    } else {
        return 4; // Obese
    }
```

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main.c... Output

```
BMI: 22.8571
Category: Normal weight


=== Code Execution Successful ===
```